

Commentary on “Comparative efficacy and safety of PSCA CAR-engineered V δ 1 $\gamma\delta$ T cells for immunotherapy of pancreatic cancer”

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To cite: de Bont DA, Straetemans T, Sebestyén Z, *et al.* Commentary on “Comparative efficacy and safety of PSCA CAR-engineered V δ 1 $\gamma\delta$ T cells for immunotherapy of pancreatic cancer”. *Journal for ImmunoTherapy of Cancer* 2026;**14**:e014199. doi:10.1136/jitc-2025-014199

► Additional supplemental material is published online only. To view, please visit the journal online (<https://doi.org/10.1136/jitc-2025-014199>).

Accepted 01 February 2026

ABSTRACT

Adoptive transfer of unmodified $\gamma\delta$ T cells has shown limited clinical benefit, prompting a shift toward chimeric antigen receptor (CAR) engineering to enhance activation, persistence, and tumor specificity. CAR technology positions $\gamma\delta$ T cells as promising carriers due to their innate-like cytotoxicity and HLA-independent recognition. Evidence from $\alpha\beta$ CAR T-cell trials indicates that $\gamma\delta$ CAR T cells can persist for months to years post-infusion, suggesting their potential contribution to long-term immune surveillance against cancer. Among $\gamma\delta$ subsets, V γ 9V δ 2 T cells dominate peripheral blood and have been preferentially used for CAR engineering. Recent advances, however, enable the expansion of V δ 1 T cells, known for their tissue residency and resistance to exhaustion. In a key comparative study, Li *et al* benchmarked PSCA-targeted CARs in V δ 1, V γ 9V δ 2, and $\alpha\beta$ T cells, showing similar short-term tumor control but distinct transcriptional and phenotypic programs. These findings highlight the need for subset-specific optimization of $\gamma\delta$ CAR-T therapies. Early studies in hematologic malignancies could provide a practical proof of concept, before tackling the additional challenges of solid tumors. Such focused development may be essential to advance the $\gamma\delta$ CAR-T field amid tightening industry investment.

Given the limited clinical activity observed with adoptively transferred natural $\gamma\delta$ T cells, the field has shifted toward engineering strategies that provide both additional activation signals and tumor antigen specificity through CARs.¹ The introduction of CAR technology has emerged as a promising strategy to enhance $\gamma\delta$ T-cell potency and redirect their activity toward tumor cells, thereby transforming them into highly effective antitumor effectors.^{1 2} Interestingly, accumulating evidence suggests that $\gamma\delta$ T cells may contribute to long-term CAR-T cell persistence when primary $\alpha\beta$ T cells are engineered without prior selection for CD4⁺ or CD8⁺ subsets. Given that peripheral blood contains 1–10% $\gamma\delta$ T cells, predominantly of the V γ 9V δ 2 subtype, these cells become transduced during manufacturing and potentially

persist after infusion. Indeed, two clinical studies of conventional $\alpha\beta$ T cell-derived CAR-T products have reported $\gamma\delta$ CAR-T cells persisting from months to years post-infusion. In one anti-CD19 CAR-T trial, $\gamma\delta$ CAR-T cells were detected both early (2.5 months) and at long-term follow-up (7.2 years), expressing cytotoxic markers such as granzyme B, CD57, and T-bet, consistent with sustained effector function.³ A similar finding was reported in another pediatric anti-CD19 CAR-T study, where $\gamma\delta$ CAR-T cells were observed at early post-infusion time points in a subset of patients.⁴ Although the clinical relevance of these populations remains to be fully understood, their persistence raises important questions about their contribution to therapeutic outcomes, and has fueled growing interest in $\gamma\delta$ T cells as carriers for CAR constructs, positioning them as promising candidates for next-generation allogeneic CAR-T cell products.¹ Their HLA-independent antigen recognition and innate-like cytotoxicity¹ underscore their potential as an “off-the-shelf” cell therapy platform, with a reduced risk of graft-versus-host disease, although some $\gamma\delta$ T cells have been reported to exhibit allo-HLA reactivity.⁵ Moreover, $\gamma\delta$ T-cell infiltration into tumors has been associated with improved prognosis across multiple cancer types.⁶ However, it is important to acknowledge that sustained persistence remains a major challenge for allogeneic CAR-T cell therapies, largely due to host-versus-graft immune responses that do not occur in autologous settings.

Adoptive transfer of unmodified $\gamma\delta$ T cells has historically relied on expansion protocols that start with unselected $\gamma\delta$ T cells, and use bisphosphonates to promote proliferation. Given the higher frequency of V γ 9V δ 2 $\gamma\delta$ T cells in peripheral blood, and the nature of stimulating agents during most expansion protocols,¹ these approaches typically result



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in a predominant expansion of the V γ 9V δ 2 subset. Consequently, most CAR-engineering studies progressing to clinic have used highly enriched V γ 9V δ 2 $\gamma\delta$ T cells as carriers (table 1). The heterogeneous composition of bulk $\gamma\delta$ T-cell products can, however, lead to variability in function, persistence, and manufacturing consistency,⁷ whereas subset-defined approaches aim to improve product standardization and predictability. Therefore, some studies have chosen to purify V γ 9V δ 2 $\gamma\delta$ T cells before engineering. In addition, more recently, the development of dedicated expansion technologies for V δ 1 $\gamma\delta$ T cells has enabled the extension of CAR engineering strategies to this subset.⁸ This is also reflected by the relatively small number of clinical studies using V δ 1 $\gamma\delta$ T cells as carriers so far. To date, however, only very limited head-to-head comparisons have been conducted^{9,10} to determine which $\gamma\delta$ T-cell carrier type may offer a higher likelihood of clinical success, or how these compare to $\alpha\beta$ T cells, which are still the gold standard carriers, due to their clinical success. Within this context, the study by Li *et al* provides important new insights by performing in vivo benchmarking of V δ 1, V γ 9V δ 2, and conventional $\alpha\beta$ T cells engineered with a PSCA-targeted CAR. The results demonstrated comparable tumor control in a pancreatic cancer model, indicating that, at least in terms of short-term tumor control, all carrier types exhibit equivalent potency.¹¹ These findings further imply that in studies reporting long-term tumor control after engineering $\alpha\beta$ T cells, where CAR-transduced $\gamma\delta$ T cells were also detected at low frequency before infusion,^{3,4} the persisting V δ 1 CAR-T cells may not merely represent residual, functionally inert cells that survived the treatment process. Rather, they might have been actively involved in the initial anti-tumor response, although at low frequency, and subsequently maintained as long-lived effector or memory-like populations. This notion is further supported by studies in disease-free survivors of acute leukemia following allogeneic bone marrow transplantation, where increased frequencies of $\gamma\delta$ T cells have been observed (reviewed in studies by Beringer *et al* and Janssen *et al*^{1,2}). These findings highlight the potential role of $\gamma\delta$ T cells in sustaining long-term immune persistence, making them attractive candidates for CAR-T cell therapy.

The capacity of engineered T cells to persist long-term as guardians against cancer may be important not only in diseases such as acute lymphoblastic leukemia, but also in more complex malignancies characterized by high tumor heterogeneity and a strongly immunosuppressive micro-environment, such as multiple myeloma. In these settings, early loss of circulating CAR-T cells has been associated with a significantly increased risk of relapse, underscoring the central role of CAR-T cell persistence in maintaining therapeutic efficacy.¹² However, the concept of memory $\gamma\delta$ T cells is less well explored compared with $\alpha\beta$ T cells, partly because many circulating human $\gamma\delta$ T cells already exhibit an effector, or terminally differentiated phenotype.^{1,2} Single-cell transcriptomic studies provided further

evidence that engineered memory cells may be important for improving clinical outcomes. For example, single-cell longitudinal analyses have highlighted transcriptional signatures associated with persistent CAR-T cells in multiple myeloma, revealing that long-lived CAR-T cells display memory-like transcriptional features, increased clonality, and enhanced metabolic fitness. Moreover, these persistent populations exhibit reduced expression of exhaustion-related genes¹³ and modulation of co-stimulatory signaling has shown that transcriptional states can be reshaped and linked to distinct functional outcomes in mouse models.¹⁴ The study by Li *et al* now adds comparative data for PSCA CAR-engineered V δ 1 and V γ 9V δ 2 $\gamma\delta$ T cells, revealing distinct transcriptional programs. The authors observed a more balanced immunophenotype for V δ 1 $\gamma\delta$ T cells, characterized by strong expression of cytotoxic markers, downregulation of immune checkpoint molecules (eg, programmed cell death protein-1), and retention of memory subsets, consistent with a less differentiated and less exhausted phenotype. However, this did not translate into a significantly increased efficacy of V δ 1 CAR-T cells in mouse models,¹¹ raising the question of whether these transcriptional differences, are truly relevant for in vivo behavior. Nonetheless, mouse models cannot fully recapitulate the complex tumor microenvironment or the dynamics of long-term memory formation observed in humans. Moreover, the type of mouse model used can substantially influence experimental outcomes. For example, in a study of MUC1-Tn-targeting V γ 9V δ 2 CAR-T cells, the killing capacity against breast cancer cell lines was similar between conventional $\alpha\beta$ CAR-T cells and V γ 9V δ 2 CAR-T cells, while the V γ 9V δ 2 CAR-T cells were more potent against gastric tumor cell lines. Yet, $\alpha\beta$ CAR-T cells demonstrated superior persistence on in vitro restimulation.⁹

Together, these studies indicate that V δ 1 CAR-T cells often display a more favorable balance between effector function, persistence, and resistance to exhaustion, whereas V γ 9V δ 2 CAR-T cells achieve rapid cytotoxic responses, but may require additional support to sustain activity. These differences highlight the need to analyze V δ 1 and V γ 9V δ 2 CAR-T cells separately, rather than treating $\gamma\delta$ T cells as a uniform population. Understanding their distinct biology is essential for optimizing $\gamma\delta$ CAR-T cell design and translation, for instance, favoring V δ 1 $\gamma\delta$ T cells for solid tumors where persistence and infiltration are critical, and V γ 9V δ 2 $\gamma\delta$ T cells for hematologic malignancies where rapid expansion is advantageous. However, compared with bulk $\gamma\delta$ T-cell products, such subset-specific approaches introduce practical challenges, including increased manufacturing complexity, higher costs, and regulatory hurdles. Moreover, intrinsic differences in starting frequency, expansion kinetics, and transduction efficiency complicate direct potency comparisons. V δ 1 $\gamma\delta$ T cells may be particularly advantageous in autologous settings (eg, acute lymphoblastic leukemia), where long-term persistence is critical for tumor eradication.³ In contrast, both V δ 1 and

Table 1 Preclinical and clinical studies exploring $\gamma\delta$ CAR-T cell therapy. The studies in green explore $\gamma\delta$ CAR-T cell therapy in hematological malignancies, those in red explore solid tumors, and those in blue explore $\gamma\delta$ CAR-T cell therapy in both

Cell source	Hematological or solid malignancy	Mode of action/ target	Co-administration	Phase	Name/program	Clinical identifier
Highly enriched Vγ9Vδ2 $\gamma\delta$ T cell						
Allogeneic	Relapsed or refractory B-cell malignancies	Anti-CD19 CAR	NA	Phase 1	UTAA09	NCT06092047 (ref. ¹⁸)
Allogeneic	Relapsed or refractory multiple myeloma	Anti-BCMA CAR	NA	Phase 1 (recruiting)	UTAA17	NCT06279026 (ref. ¹⁹)
Allogeneic	Relapsed or refractory B-cell acute lymphoblastic leukemia	Anti-CD19 CAR	NA	Phase 1 (recruiting)	QH103	NCT06056752 (ref. ²⁰)
Allogeneic	Relapsed or refractory B-cell non-Hodgkin's lymphoma	Anti-CD19 CAR	NA	Phase 1/2 (recruiting)	CHN-PLAGH-BT-072	NCT05554939 (ref. ²¹)
Allogeneic	Relapsed or refractory B-cell non-Hodgkin's lymphoma	Bispecific anti-CD19 and anti-CD20 CAR	NA	Phase 1 (recruiting)	LUCAR-G39D	NCT06395870 (ref. ²²)
Allogeneic	Relapsed or refractory T-cell-derived malignancies	Anti-CD7 CAR	NA	Phase 1	PA3-17	NCT05170568 (ref. ²³)
Allogeneic	Relapsed or refractory AML	Anti-CD123 CAR	NA	Phase 1	CAR- $\gamma\delta$ T cell for AML	NCT04796441/ NCT05388305 (ref. ^{24,25})
Allogeneic	Relapsed or refractory T-cell acute lymphoblastic leukemia	Anti-CD38 CAR	NA	Preclinical	Anti-CD38KO/CD38-CAR $\gamma\delta$ T cells	NA (ref. ²⁶)
Allogeneic	Advanced hematological or solid malignancies	Anti-NKG2D ligand CAR	NA	Phase 1 (recruiting /early stage)	CTM-N2D	NCT05302037 (ref. ²⁷)
Allogeneic	Advanced malignant solid tumors	Anti-NKG2D ligand CAR	NA	Phase 1 (completed)	UTAA06	NCT06372236 (ref. ²⁸)
Allogeneic	Relapsed/refractory triple-negative breast cancer, non-small cell lung cancer, colorectal cancer, glioblastoma	Anti-HLA-G CAR	NA	Phase 1/2 (recruiting)	CAR001	NCT06150885 (ref. ²⁹)
Allogeneic/haploidentical	Relapsed or refractory solid tumors	Anti-NKG2D ligand CAR	NA	Phase 1	CTM-N2D-101	NCT04107142 (ref. ³⁰)
Allogeneic	Malignant brain glioma and meningeal metastasis	Anti-B7H3 CAR	NA	Phase 1/2 (recruiting)	QH104	NCT06018363/ NCT06592092 (ref. ^{31,32})
Autologous	GPC3/mesothelin expressing solid tumors	Anti-GPC3/mesothelin CAR	PD-1/PDL-1/ CTLA-4 antibodies	Phase 1 (recruiting)	GPC3/Mesothelin-CAR	NCT06196294 (ref. ³³)
Autologous	End-stage metastatic castration-resistant prostate cancer	Anti-PSCA $\gamma\delta$ 8t28Z CAR	NA	Phase 1 (recruiting)	MSGV1-PSCA-8T28Z	NCT06193486 (ref. ³⁴)
Allogeneic	Multiple myeloma	Anti-folate receptor- α CAR	NA	Preclinical	Anti-CD38-CAR $\gamma\delta$ T cells	NA (ref. ³⁵)

Continued

Table 1 Continued						
Cell source	Hematological or solid malignancy	Mode of action/ target	Co-administration	Phase	Name/program	Clinical identifier
Allogeneic	Triple-negative breast cancer	Anti-MCSP CAR	NA	Preclinical	7×19 CAR-γδTγδ	NA (ref. ³⁶)
Allogeneic	Melanoma	Anti-CD123 CAR	NA	Preclinical	MCSP-CAR	NA (ref. ³⁷)
Allogeneic	Neuroblastoma	Anti-GD2 CAR	NA	Preclinical	GD2 CAR γδ (Vδ1/Vδ2)	NA (ref. ¹⁰)
Vδ2 γδ T cells						
Allogeneic	B-cell malignancies	Anti-CD20 CAR	NA	Phase 1 (terminated)	ADI-001	NCT04735471 (ref. ³⁸)
Allogeneic	Neuroblastoma	Anti-GD2 CAR	NA	Preclinical	GD2-DAP10 CAR	NA (ref. ^{39 40})
Allogeneic	Solid tumors	Anti-MUC1-Tn CAR	NA	Preclinical	MUC1-Tn CAR γδ T cells	NA (ref. ⁹)
Allogeneic	NKG2D ligand positive tumors	Anti-NKG2D CAR	NA	Preclinical	NKG2Dz-CAR	NA (ref. ⁴¹)
Vδ1 γδ T cells						
Allogeneic	Relapsed or refractory AML	Anti-CD123 CAR	NA	Preclinical	CD123CAR-DOT	NA (ref. ⁸)
Allogeneic	Relapsed or renal cell carcinoma	Anti-CD70 CAR	NA	Phase 1/2	ADI-270	NCT06480565 (ref. ⁴²)
Allogeneic	Hepatocellular carcinoma	Anti-glypican-3 CAR+sLL15	NA	Preclinical	ADI-200	NA (ref. ⁴³)
Allogeneic	Multiple solid tumors	Anti-B7H3 CAR	IL-2 co-expression	Preclinical	B7-H3-CAR Vδ1	NA (ref. ⁴⁴)
AML, acute myeloid leukemia; CAR, chimeric antigen receptor; CTLA-4, cytotoxic T-lymphocytes-associated protein 4; IL, interleukin; PD-1, programmed cell death protein-1; PD-L1, programmed death-ligand 1.						

V γ 9V δ 2 subsets may be suitable for allogeneic products, with subset-specific homing guiding tumor selection. For example, glioblastoma is enriched for V γ 9V δ 2 $\gamma\delta$ T cells¹⁵ and might benefit from using engineered V γ 9V δ 2 $\gamma\delta$ T cells, whereas other solid tumors, such as prostate cancer, are predominantly infiltrated by V δ 1 $\gamma\delta$ T cells¹ and may preferentially benefit from engineered V δ 1 $\gamma\delta$ T cells as reported by Li *et al.*¹¹

For clinical development, the relevance of long-term persistence may be particularly important when autologous products are used. Whether this benefit also extends to allogeneic “off-the-shelf” products remains questionable, as these are typically rapidly rejected by the host immune system. In allogeneic “off-the-shelf” settings, where $\gamma\delta$ CAR-T cells are intended to provide transient antitumor activity, limited persistence may not be disadvantageous. Instead, short-lived but potent responses could enhance safety and enable redosing strategies. Another important consideration is the disease context for early clinical testing. Demonstrating efficacy in hematologic malignancies or autoimmune diseases could offer a feasible proof of concept for $\gamma\delta$ CAR T therapy, either autologous or allogeneic, before addressing the additional challenges of solid tumors, such as antigen heterogeneity, immune exclusion, and an immunosuppressive microenvironment. Establishing success in a controlled setting with $\alpha\beta$ T cells as proven carriers, particularly those engineered for allogeneic use with $\alpha\beta$ T cell receptor (TCR) knockout,¹⁶ may therefore accelerate confidence, clinical acceptance, and investment in this platform. Given the current financial headwinds in the engineering T-cell therapy field, illustrated by the withdrawal of several industrial pipelines, including Takeda’s discontinuation of its $\gamma\delta$ T-cell therapy program (<https://www.oncologypipeline.com/apexonco/takedas-cancer-prospects-dim-further>), prioritizing lower-risk, high-impact indications may be critical to sustaining momentum, and validating the platform’s clinical potential. This consideration is especially important if academic pharma needs to take the first successful steps.¹⁷

In conclusion, the comparative preclinical study by Li *et al.*¹¹ represents an important step forward for $\gamma\delta$ CAR-T cell development. By directly comparing V δ 1, V γ 9V δ 2, and $\alpha\beta$ T-cell carriers, it provides valuable insight into subset-specific strengths and limitations that bulk $\gamma\delta$ T-cell approaches cannot reveal. Extending such comparative studies into both hematologic and solid tumor models, and ultimately into early-phase clinical trials, will be key to defining the optimal indications, persistence profiles, and manufacturing strategies for next-generation $\gamma\delta$ CAR-T therapies.

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Contributors DAdB and JK wrote and reviewed the original draft. ZS and TS edited and reviewed. AI was used for language checking.

Funding Funding for this commentary was provided by Oncode Accelerator (Oncode Accelerator | Advancing Cancer Drug Development).

Competing interests JK is a shareholder of Gadeta. JK and ZS are inventors on patents with $\gamma\delta$ TCR related topics. JK and ZS are inventors on patents with CD277-related topics.

Patient consent for publication Not applicable.

Ethics approval Not applicable.

Provenance and peer review Commissioned; externally peer reviewed.

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REFERENCES

- Beringer DX, Straetmans T, Minguet S, *et al.* Disrupting the balance between activating and inhibitory receptors of $\gamma\delta$ T cells for effective cancer immunotherapy. *Nat Rev Cancer* 2025;25:590–612.
- Janssen A, Villacorta Hidalgo J, Beringer DX, *et al.* $\gamma\delta$ T-cell Receptors Derived from Breast Cancer-Infiltrating T Lymphocytes Mediate Antitumor Reactivity. *Cancer Immunol Res* 2020;8:530–43.
- Melenhorst JJ, Chen GM, Wang M, *et al.* Decade-long leukaemia remissions with persistence of CD4+ CAR T cells. *Nature New Biol* 2022;602:503–9.
- Anderson ND, Birch J, Accogli T, *et al.* Transcriptional signatures associated with persisting CD19 CAR-T cells in children with leukemia. *Nat Med* 2023;29:1700–9.
- GJJ K, W S, AD M, *et al.* Identification of a tumor-specific allo-HLA-restricted $\gamma\delta$ TCR - PubMed. *Blood Adv* 2019;3.
- Gentles AJ, Newman AM, Liu CL, *et al.* The prognostic landscape of genes and infiltrating immune cells across human cancers. *Nat Med* 2015;21:938–45.
- Wang CQ, Lim PY, Tan AH-M. Gamma/delta T cells as cellular vehicles for anti-tumor immunity. *Front Immunol* 2023;14.
- Sánchez Martínez D, Tirado N, Mensurado S, *et al.* Generation and proof-of-concept for allogeneic CD123 CAR-Delta One T (DOT) cells in acute myeloid leukemia. *J Immunother Cancer* 2022;10:e005400.
- Zhai X, You F, Xiang S, *et al.* MUC1-Tn-targeting chimeric antigen receptor-modified V γ 9V δ 2 T cells with enhanced antigen-specific anti-tumor activity. *Am J Cancer Res* 2021;11:79–91.
- Capsomidis A, Benthall G, Van Acker HH, *et al.* Chimeric Antigen Receptor-Engineered Human Gamma Delta T Cells: Enhanced Cytotoxicity with Retention of Cross Presentation. *Mol Ther* 2018;26:354–65.
- Li Z, Liu N, Shah Z, *et al.* Comparative efficacy and safety of PSICA CAR-engineered V δ 1 $\gamma\delta$ T cells for immunotherapy of pancreatic cancer. *J Immunother Cancer* 2025;13:e011890.
- Ledergor G, Fan Z, Wu K, *et al.* CD4+ CAR T-cell exhaustion associated with early relapse of multiple myeloma after BCMA CAR T-cell therapy. *Blood Adv* 2024;8:3562–75.
- A longitudinal single-cell atlas to predict outcome and toxicity after bcma-directed car t cell therapy in multiple myeloma. 2025.
- Hernández-López P, van Diest E, Brazda P, *et al.* Dual targeting of cancer metabolome and stress antigens affects transcriptomic heterogeneity and efficacy of engineered T cells. *Nat Immunol* 2024;25:88–101.
- Nicolasen MJT, Gatti LC, Gasull-Celades L, *et al.* Autologous profiling reveals inter-patient heterogeneity in V δ 2 + $\gamma\delta$ TIL responses to glioblastoma driven by extracellular matrix-BTN3A axis. *J Immunother Cancer* 2025;13.
- Mailankody S, Matous JV, Chhabra S, *et al.* Allogeneic BCMA-targeting CAR T cells in relapsed/refractory multiple myeloma: phase 1 UNIVERSAL trial interim results. *Nat Med* 2023;29:422–9.

- 17 Johanna I, Daudeij A, Devina F, *et al.* Basics of advanced therapy medicinal product development in academic pharma and the role of a GMP simulation unit. *Immuno-Oncology and Technology* 2023;20.
- 18 UTAA09 injection in the treatment of relapsed/refractory hematolymphatic malignancies. In. ClinicalTrials.gov; 2023.
- 19 NCT06279026. UTAA17 injection in the treatment of relapsed/refractory multiple myeloma. In. ClinicalTrials.gov; 2024.
- 20 NCT06056752. QH103 cell injection for the treatment of relapsed/refractory b-cell acute lymphoblastic leukemia. In. ClinicalTrials.gov; 2023.
- 21 NCT05554939. Allogenic cd19-targeting car- $\gamma\delta$ cell therapy in r/r nhl. In. ClinicalTrials.gov; 2022.
- 22 NCT06395870. Targeting cd19/cd20 dual-targeted cell in patients with relapsed/refractory b-cell non-hodgkin lymphoma. In. ClinicalTrials.gov; 2024.
- 23 NCT05170568. PA3-17 injection treatment of adult patients with cd7-positive relapsed/refractory lymphoid hematologic malignancies. In. ClinicalTrials.gov; 2021.
- 24 NCT05388305. Universal car- $\gamma\delta$ cell injection in the aml patients. In. ClinicalTrials.gov; 2022.
- 25 NCT04796441. Clinical study of universal car- $\gamma\delta$ cell injection in the treatment of patients with relapsed aml after transplantation. In. ClinicalTrialsGov; 2020.
- 26 Snyder G, Martin AK, Sezgin Y, *et al.* Off-the-shelf allogeneic polyclonal cd38ko/cd38-car $\gamma\delta$ cells for the treatment of t cell malignancies. *Immunology* 2025.
- 27 NCT05302037. Allogeneic nkg2dl-targeting car $\gamma\delta$ t cells (ctm-n2d) in advanced cancers(ANGELICA). In. ClinicalTrials.gov; 2021.
- 28 NCT06372236. UTAA06 injection for treatment of advanced malignant solid tumors. In ClinicalTrialsGov; 2023.
- 29 NCT06150885. A safety and efficacy study of allogeneic car gamma-delta t cells in subjects with relapsed/refractory solid tumors (car001). In. ClinicalTrials.gov; 2024.
- 30 NCT04107142. Haplo / allogeneic nkg2dl-targeting chimeric antigen receptor-grafted $\gamma\delta$ t cells for relapsed or refractory solid tumour. In ClinicalTrialsGov; 2019.
- 31 NCT06018363. Linical study on the treatment of malignant brain glioma by qh104 cell injection. In. ClinicalTrials.gov; 2023.
- 32 NCT06592092. Clinical study of qh104 cell injection for the treatment of meningeal metastases of b7h3+ solid tumors. In. ClinicalTrials.gov.
- 33 NCT06196294. GPC3/mesothelin-car- $\gamma\delta$ cells against cancers. In. ClinicalTrials.gov; 2024.
- 34 NCT06193486. Autologous gamma delta t cells to target prostate stem cell antigen in mcrpc. In. ClinicalTrials.gov; 2024.
- 35 Hattori Y, Ohshima N, Taniguchi Y, *et al.* 22O Development of anti-CD38-CAR allogeneic $\gamma\delta$ T cells as an off-the-shelf use for myeloma. *ESMO Open* 2025;10:104179.
- 36 Ye X, Deng X, Wen J, *et al.* Folate Receptor-Alpha Targeted 7x19 CAR- $\gamma\delta$ T Suppressed Triple-Negative Breast Cancer Xenograft Model in Mice. *J Oncol* 2022;2022.
- 37 Harrer DC, Simon B, Fujii S-I, *et al.* RNA-transfection of $\gamma\delta$ T cells with a chimeric antigen receptor or an $\alpha\beta$ T-cell receptor: a safer alternative to genetically engineered $\alpha\beta$ T cells for the immunotherapy of melanoma. *BMC Cancer* 2017;17:551.
- 38 Neelapu SS, Stevens DA, Hamadani M, *et al.* A Phase 1 Study of ADI-001: Anti-CD20 CAR-Engineered Allogeneic Gamma Delta1 ($\gamma\delta$) T Cells in Adults with B-Cell Malignancies. *Blood* 2022;140:4617-9.
- 39 Ganapathy T, Radhakrishnan R, Sakshi S, *et al.* CAR $\gamma\delta$ T cells for cancer immunotherapy. Is the field more yellow than green? *Cancer Immunol Immunother* 2023;72:277-86.
- 40 Fisher J, Abramowski P, Wisidagamage Don ND, *et al.* Avoidance of On-Target Off-Tumor Activation Using a Co-stimulation-Only Chimeric Antigen Receptor. *Mol Ther* 2017;25:1234-47.
- 41 Ang WX, Ng YY, Xiao L, *et al.* Electroporation of NKG2D RNA CAR Improves V γ 9V δ 2 T Cell Responses against Human Solid Tumor Xenografts. *Mol Ther Oncolytics* 2020;17:421-30.
- 42 NCT06480565. A phase 1/2 trial of adi-270 in ccrcc. In ClinicalTrialsGov; 2024.
- 43 Makkouk A, Yang XC, Barca T, *et al.* Off-the-shelf V δ 1 gamma delta T cells engineered with glypican-3 (GPC-3)-specific chimeric antigen receptor (CAR) and soluble IL-15 display robust antitumor efficacy against hepatocellular carcinoma. *J Immunother Cancer* 2021;9.
- 44 Jiang L, You F, Wu H, *et al.* B7-H3-Targeted CAR-V δ 1T Cells Exhibit Potent Broad-Spectrum Activity against Solid Tumors. *Cancer Res* 2024;84:23:4066-80:.